

TANGENT GALVANOMETER (09)

Aim

To determine the horizontal component of the Earth's magnetic field (B_0), the internal resistance (G) of the galvanometer and the magnetic field (B) of a bar magnet using tangent galvanometer.

Apparatus

Tangent galvanometer, variable resistor, reversing key, spirit level, power supply (5 V), bar magnet, connecting wires and meter ruler.

Theory

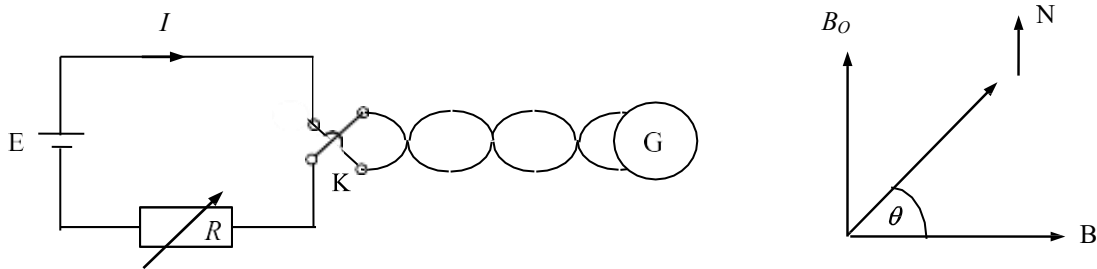


Fig. 1: The figure of the experimental circuit and the orientation of the magnetic fields
Where, I : current through the circuit, E : e.m.f. of the power supply, G : internal resistance of the tangent galvanometer, R : resistance of the variable resistor, K : reversing key, B_0 : horizontal component of the Earth's magnetic fields, B : magnetic field due to I

From Kirchhoff's law,

$$I = \frac{E}{(R+G)} \quad (1)$$

When a current I pass through the coil, magnetic field at the center of the coil (B) is given by,

$$B = \frac{\mu_o n I}{2r} \quad (2)$$

where,

μ_o : permeability of free space

n : number of turns of the coil

r : radius of the coil

When the plane of the galvanometer coil is set such that B is perpendicular to B_0 ,

$$B_0 = B \tan \theta \quad (3)$$

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From (1), (2) & (3),

$$\begin{aligned}\frac{B_o}{\tan\theta} &= \frac{\mu_o n E}{2r(R + G)} \\ \tan\theta &= \frac{2r B_o}{\mu_o n E} R + \frac{2r B_o G}{\mu_o n E}\end{aligned}\quad (4)$$

From the graph of $\tan \theta$ vs R , B_o and G can be found.

Procedure

- Level and set the tangent galvanometer so that the plane of the coil lies along the magnetic N-S direction and the needle reads zero.
- Connect the circuit as shown in the Fig. 1.
- Set the value of R to zero and take relevant deflections on the scale.
- Reverse the key (K) and take the relevant readings.

Q: What is the purpose of reversing the key (K)?

- Repeat the above procedure for another few values of R to find B_o using a graphical method.

Q: Is it possible to take deflection readings for 100 different R values?

- Place the bar magnet on the tangent galvanometer, so that the bar magnet is parallel to the coil of the tangent galvanometer. (In this position needle should be directed with zero mark.)
- Set the value of R to 40Ω and take relevant deflections on the scale.

Q: What is the reason for selecting R as 40Ω in this part of the experiment?

- Change the magnetic N-S direction of the bar magnet (In this position needle should be directed with zero mark) and take the relevant deflections.
- Reverse the key (K) and take the relevant readings.
- Take relevant measurement to calculate the radius of the coil (r) and note down the number of turns of the coil (n) and e.m.f. of the power supply (E).